

EDUCATION

Duke University	2020 – present
PhD Candidate, Earth and Climate Sciences	
Advisor: Avner Vengosh	
University of California, Santa Barbara	2015 – 2019
B.S. Earth Science, Emphasis in Geohydrology	

PUBLICATIONS

(*co-first author, †undergraduate mentee)

[Google Scholar Profile](#)

In Preparation (In Manuscript Form)

- [3] Lu, P. L.; Harmon, R.S.; Leatherman, L.; **Williams, G.D.Z.**; Curry, A.C.; Kelly, L.; Dwyer, G.S.; Reynolds, V.S.; Dossou, A.; Richter, D.D., Behavior of Lithium during soil formation in pegmatitic systems: Characterizing soil lithium signatures over and around Li-pegmatites
- [2] Nativ, P.; **Williams, G.D.Z.**; Vengosh, A., From pond to process: a reliable method for pH measurements in hypersaline solutions
- [1] Wudke, H.; **Williams, G.D.Z.**; Vengosh, A., Baseline Water Quality of Shallow Groundwater used by Indigenous Communities near the Salar de Uyuni, Bolivia

Submitted or In-Review

- [4] †Schaffer, A.; **Williams, G.D.Z.**; Dwyer, G.S.; Hyland, E.G.; Wong, J.M.; Osterberg, J.S.; Schultz, T.F.; Kipp, M.A., *Submitted to Geochimica et Cosmochimica Acta*. Environmental and vital effects on lithium isotopes and trace element ratios in biogenic carbonates: Insights from an estuarine salinity gradient
- [3] †Hall, G.A.; **Williams, G.D.Z.**; Friedrich, A.J.; Vengosh, A., *Submitted to Applied Geochemistry*. The Potential Water Quality Impacts of Mining Critical Minerals: Comparative Analysis of Simulated Leachates of Sulphide and Laterite Ores
- [2] **Williams, G.D.Z.**; Barre, J.; Louvat, P.; Bérail, S.; Millot, R.; Vengosh, A., *In-Revision at Earth and Planetary Science Letters*. Geochemical controls on the formation of lithium brines in closed-basins of the Lithium Triangle
- [1] Nativ, P.; **Williams, G.D.Z.**; Vengosh, A. *In-Revision at Environmental Science & Technology Letters*. Discrepances Between pH and Corrosive Indexes of Hypersaline Effluents

Published

- [13] **Williams, G.D.Z.**; Petrović, M.; Hill, R.C.; †Hall, G.A.; Vengosh, A., 2025. The Water Quality Impacts of Legacy Hard-Rock Lithium Mining and Processing. *Environmental Science & Technology* 59, 26492–26505. <https://doi.org/10.1021/acs.est.5c13682>
- [12] *†Hall, G.A.; ***Williams, G.D.Z.**; Sirbescu, M.L.C.; Lu, P.L.; Dwyer, G.S.; Richter, D.D.; Vengosh, A., 2025. Strontium isotopes and Rb/Sr tracers in surface soils for locating subsurface lithium pegmatites. *Applied Geochemistry* 195, 106631. <https://doi.org/10.1016/j.apgeochem.2025.106631>
- [11] Hill, R.C.; Wang, Z.; Hu, J.; **Williams, G.D.Z.**; Vengosh, A., 2025. Radionuclides and the uranium isotope fingerprint of globally produced phosphate rocks, mineral fertilizers, and phosphogypsum waste and its potential effect on the environment. *Journal of Hazardous Materials* 499, 140033. <https://doi.org/10.1016/j.jhazmat.2025.140033>
- [10] Lopez, R.F.; Huayta, J.; **Williams, G.D.Z.**; Seay, S.A.; Lalwani, P.D.; Bacot, S.; Vengosh, A.; Meyer, J., 2025. Lithium Nickel Manganese Cobalt Oxide Particles Cause Developmental Neurotoxicity in *Caenorhabditis elegans*. *Environmental Science: Advances*. <https://doi.org/10.1039/D5VA00103J>

- [9] **Williams, G.D.Z.**; Nativ, P.; Vengosh, A., 2025. The role of boron in controlling the pH of lithium brines. *Science Advances* 11, eadw3268. <https://doi.org/10.1126/sciadv.adw3268>
- [8] **Williams, G.D.Z.**; Vengosh, A., 2025. Quality of Wastewater from Lithium-Brine Mining. *Environmental Science & Technology Letters* 12, 151–157. <https://doi.org/10.1021/acs.estlett.4c01124>
- [7] **Williams, G.D.Z.**; Saltman, S.; Wang, Z.; Warren, D.M.; Hill, R.C.; Vengosh, A., 2024. The potential water quality impacts of hard-rock lithium mining: Insights from a legacy pegmatite mine in North Carolina, USA. *Science of The Total Environment* 956, 177281. <https://doi.org/10.1016/j.scitotenv.2024.177281>
- [6] Hill, R.C.; Wang, Z.; **Williams, G.D.Z.**; Polyak, V.; Singh, A.; Kipp, M.A.; Asmerom, Y.; Vengosh, A., 2024. Reconstructing the depositional environment and diagenetic modification of global phosphate deposits through integration of uranium and strontium isotopes. *Chemical Geology* 662, 122214. <https://doi.org/10.1016/j.chemgeo.2024.122214>
- [5] Hill, R.C.; **Williams, G.D.Z.**; Wang, Z.; Hu, J.; El-Hasan, T.; Duckworth, O.W.; Schnug, E.; Bol, R.; Singh, A.; Vengosh, A., 2024. Tracing the Environmental Effects of Mineral Fertilizer Application with Trace Elements and Strontium Isotope Variations. *Environmental Science & Technology Letters* 11, 604–610 <https://doi.org/10.1021/acs.estlett.4c00170>
- [4] Hu, J.; Wang, Z.; **Williams, G.D.Z.**; Dwyer, G.S.; Gatiboni, L.; Duckworth, O.W.; Vengosh, A., 2024. Evidence for the accumulation of toxic metal(loid)s in agricultural soils impacted from long-term application of phosphate fertilizer. *Science of The Total Environment* 907, 167863. <https://doi.org/10.1016/j.scitotenv.2023.167863>
- [3] Wang, Z.; Hill, R.; **Williams, G.**; Dwyer, G.S.; Hu, J.; Schnug, E.; Bol, R.; Sun, Y.; Coleman, D.S.; Liu, X.-M.; Sandstrom, M.R.; Vengosh, A., 2023. Lead isotopes and rare earth elements geochemistry of global phosphate rocks: Insights into depositional conditions and environmental tracing. *Chemical Geology* 639, 121715. <https://doi.org/10.1016/j.chemgeo.2023.121715>
- [2] Vengosh, A.; Wang, Z.; **Williams, G.**; Hill, R.; M. Coyte, R.; Dwyer, G. S., 2022. The Strontium Isotope Fingerprint of Phosphate Rocks Mining. *Science of The Total Environment*, 850, 157971. <https://doi.org/10.1016/j.scitotenv.2022.157971>.
- [1] Wang, K.; Ellsworth, W. L.; Beroza, G. C.; **Williams, G.**; Zhang, M.; Schroeder, D.; Rubinstein, J., 2018, Seismology with Dark Data: Image-Based Processing of Analog Records Using Machine Learning for the Rangely Earthquake Control Experiment. *Seismological Research Letters*, 90 (2A), 553–562. <https://doi.org/10.1785/0220180298>.

INVITED PRESENTATIONS

Indiana University Bloomington, Seminar on Critical Minerals and Climate Change Mitigation, December 2025, *Lithium Mining, Processing, and Environmental Impacts*

Goldschmidt Conference Session on Lithium Resources for the Energy Transition, July 2025, *The Many Roles of Boron in Controlling Lithium-Brine Geochemistry*. Abstract by Williams, G.D.Z. & Vengosh A.

CONFERENCE PRESENTATIONS & ABSTRACTS

Williams, GDZ.; Vengosh, A. (2025) [oral] The Role of Boron in Controlling the pH of Lithium Brines. The 3rd International Association of Geochemistry Conference.

Williams, GDZ.; Hall, GA.; Hill, RC.; Petrović, M.; Louvat, P.; Berail, S.; Barre, J.; Millot, R.; Warner, NR.; Baker, P.; Vengosh, A. (2025) [oral] The Origin of Lithium at the Salar de Uyuni: A multi-isotope and elemental approach. The 3rd International Association of Geochemistry Conference.

- Williams, GDZ.**; Vengosh, A. (2025) [oral] The role of boron in controlling the pH of evaporated lithium brines. AEESP Conference
- Williams, GDZ.**; Hill, RC.; Wang, Z.; Kipp, MA.; Vengosh, A. (2024) [poster] Searching for Alternative Critical Mineral Sources. Symposium on Critical Resources, Minerals, and Materials Joint Efforts
- Williams, GDZ.**; Hall, G.; Petrović, M.; Hill, RC.; Dwyer, G.; Vengosh, A. (2024) [oral] Water quality in a legacy lithium mining district of North Carolina. Goldschmidt Conference
- Williams, GDZ.** (2024) [oral] Geology and Production of Latin American Lithium Deposits. North Carolina Conference on Latin American Studies
- Williams, G.**; Wang, Z.; Hill, R.; Vengosh, A. (2023) [oral] Potential Water Quality Impacts of Hard-Rock Lithium Mining. AGU Fall Meeting
- Williams, G.**; Wang, Z.; Hill, R.; Vengosh, A. (2023) [poster] Water Quality and Legacy Lithium Mining in North Carolina: Insights on the Impacts of Future Lithium Mining. Goldschmidt Conference
- Williams, G.**; Vengosh, A.; Wang, Z.; Hill, R. (2022) [oral] Elemental Fluxes in Global Phosphate Ores: Evaluation of a Potential Resource for Critical Elements. GSA Connects
- Williams, G.**; Hill, R.; Wang, Z.; Vengosh, A. (2022) [poster] Multiple Isotopes as Potential Tracers for Contaminants Derived from Lithium Mine Wastes. GSA Connects
- Williams, G.**; Hill, R.; Wang, Z.; Whittaker, M.; Stringfellow, W.; Vengosh, A. (2022) [poster] Strontium Isotope Geochemistry as a Potential Tracer for Contaminants Derived from Lithium Mine Wastes. Goldschmidt Conference

ACADEMIC SERVICE & EXPERIENCE

Duke University, Earth & Climate Science Department Seminar Organizer & Discussion Leader, 2022-2024
Conference Session Co-Chair

- 2026 (upcoming), Goldschmidt, Montréal, Canada; C Chen, T Martins, F Putzolu, **GDZ Williams** – *Lithium resources across Earth systems*
- 2025, The 3rd International Association of Geochemistry Conference, Cagliari, Italy; E Sacchi, **GDZ Williams** – *Isotopic Tools and Applications for Sustainable Water Management, Mining-Related Settings, Environmental Sciences, and the Energy Transition*

Reviewer: *Geochimica et Cosmochimica Acta*, Applied Geochemistry

Society Affiliations: Geochemical Society, International Association of Geochemistry, American Geophysical Union, Geological Society of America

GRANTS & AWARDS

Global Student Research Fund, Duke Office of Global Affairs (**\$2,145**), 2025

Fall Tuition Scholarship, Duke Graduate School (**\$5,215**), 2025

Summer Research Fellowship, Duke Graduate School (**\$14,850**), 2025

Chateaubriand Fellowship, French Embassy to the USA (**€7,967**), to conduct lithium isotope work in Pau, France at UPPA, 2024

Dissertation Research Travel Award: International, Duke Graduate School (**\$5,000**), 2023-2024

Dissertation Research Travel Award: Domestic, Duke Graduate School (**\$4,000**), 2023-2024

Potential Impacts of Lithium Mining on Water Quality in North Carolina, NC Water Resources Research Institute (**\$120,000 direct**), PI Avner Vengosh (written with Gordon Williams), 2023-2025

The Potential Environmental Effects of Lithium Mining and Extraction, Albemarle Corporation (**\$396,230 direct**), G. Williams, A. Vengosh, D. Shindell, 2023-2024

Graduate Student Research Grant, Geological Society of America (**\$1,749**), 2022

Student Travel Grant, Geological Society of America, 2022

TEACHING & MENTORING

Graduate Teaching Assistant

- EOS 101 Dynamic Earth** with Dr. Emily Klein, Fall 2020
- EOS 101 Dynamic Earth** with Dr. Alex Glass, Spring 2021
- EOS 220 Water Sciences** with Dr. Avner Vengosh, Fall 2021
- EOS 524 Water Quality and Health** with Dr. Avner Vengosh, Spring 2022
- ECS 401 Geology of North Carolina (with multi-day field trips)** with Dr. Alex Glass, Fall 2022
- ECS 201L Earth Materials** with Dr. Adam Curry, Spring 2023
- ECS 410 Senior Capstone Experience, The Geology of Ireland (with 10-day field trip)** with Dr. Gary Dwyer, Spring 2023 & Spring 2025

Guest Lectures & Field Trip Demonstrations/Presentations

- ECS 524, Water Quality and Health
- ECS 201L, Earth Materials
- ECS 210S, Exploring Earth Science: Field and Laboratory Investigations
- ECS 226S, Field Methods in Earth and Environmental Sciences

Mentored Master's Student Research

1. Sam Saltman, 2022-2023, Investigating the potential environmental impact of lithium mining in North Carolina
2. Ryan Parks, 2022-2023, Investigating the current and historical groundwater quality at the Maplewood Cemetery and Pauli Murray Center

Mentored Undergraduate Research

1. Alexandra Schaffer, 2025-present. project: Understanding the lithium isotope and trace metal geochemistry of oyster shells
2. Grace Hall, 2023-2025. thesis: "The Potential Water Quality Impacts of Mining Critical Raw Materials: Comparative Analysis of Simulated Leachates of Sulphide and Laterite Ores"
3. Tiana Dinham, 2023, project: Investigating the water quality impacts of hard-rock lithium mining in North Carolina
4. Carsten Pran, B.S. (2022), thesis: "An Interdisciplinary Approach to Understanding Duke University's Relationship with the Durham County Water System"
5. Bass Connections Project (2021) with an undergraduate team of 5 students: "Inspecting the Taps: Radium, Groundwater, and Public Health in the Heart of Texas"

PREVIOUS RESEARCH EXPERIENCE

Research Assistant in Professor Tiziana Vanorio's group, Department of Geophysics, Stanford University, January 2020 – April 2020 (ended due to COVID)

Research Assistant in Professor Tiziana Vanorio's group, Department of Geophysics, Stanford University, June 2019 – August 2019 (Summer Internship)

Independent Research Assistant in Professor Matthew Jackson's group, Department of Earth Sciences, University of California, Santa Barbara, February 2019 – June 2019

Research Assistant in Professor William Ellsworth's group, Department of Geophysics, Stanford University, June 2018 – August 2018 (Summer Internship)

PROFESSIONAL REFERENCES

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